

Specical topics in limsup & liminf

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1 Limit superior & Limit inferior

Proposition 2.3.1 (Basic Analysis I by Jiri Lebl)

Let $\{x_n\}_{n=1}^{\infty}$ be a bounded sequence. Define the sequences $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ by

$$a_n := \sup\{x_k : k \geq n\}, \quad b_n := \inf\{x_k : k \geq n\}.$$

Define, if the limits exist,

$$\limsup_{n \rightarrow \infty} x_n := \lim_{n \rightarrow \infty} a_n, \quad \liminf_{n \rightarrow \infty} x_n := \lim_{n \rightarrow \infty} b_n.$$

Proposition 2.3.2. (Basic Analysis I by Jiri Lebl)

Let $\{x_n\}_{n=1}^{\infty}$ be a bounded sequence. Let a_n and b_n be as in the definition above.

- (i) The sequence $\{a_n\}_{n=1}^{\infty}$ is bounded monotone decreasing and $\{b_n\}_{n=1}^{\infty}$ is bounded monotone increasing. In particular,

$$\liminf_{n \rightarrow \infty} x_n \quad \text{and} \quad \limsup_{n \rightarrow \infty} x_n \quad \text{exist.}$$

- (ii)

$$\limsup_{n \rightarrow \infty} x_n = \inf\{a_n : n \in \mathbb{N}\} \quad \text{and} \quad \liminf_{n \rightarrow \infty} x_n = \sup\{b_n : n \in \mathbb{N}\}.$$

- (iii)

$$\liminf_{n \rightarrow \infty} x_n \leq \limsup_{n \rightarrow \infty} x_n.$$

2 Limits of Sets

The definition of convergence concepts for random variables rests on manipulations of sequences of events which require limits of sets. Let $A_n \subset \Omega$. We define

$$\begin{aligned} \inf_{k \geq n} A_k &:= \bigcap_{k=n}^{\infty} A_k, & \sup_{k \geq n} A_k &:= \bigcup_{k=n}^{\infty} A_k \\ \liminf_{n \rightarrow \infty} A_n &= \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k, & \limsup_{n \rightarrow \infty} A_n &= \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k. \end{aligned}$$

The *limit* of a sequence of sets is defined as follows: If for some sequence $\{B_n\}$ of subsets

$$\limsup_{n \rightarrow \infty} B_n = \liminf_{n \rightarrow \infty} B_n = B,$$

then B is called the limit of B_n and we write $\lim_{n \rightarrow \infty} B_n = B$ or $B_n \rightarrow B$. It will be demonstrated soon that

$$\liminf_{n \rightarrow \infty} A_n = \lim_{n \rightarrow \infty} \left(\inf_{k \geq n} A_k \right)$$

and

$$\limsup_{n \rightarrow \infty} A_n = \lim_{n \rightarrow \infty} \left(\sup_{k \geq n} A_k \right).$$

Remark-1: In real numbers, it's naturally we say $a < b$. But in sets theory, if we want to compare the size of two sets A and B. We should firstly make A belongs to B or B belongs to A. For example, if $A \subset B$, then we can say B is kind of larger than A.

Remark-2: So consider a sequence of sets $\{A_k\}_{k \geq n}$, we can say the $\inf_{k \geq n} A_k$ is $\bigcap_{k=n}^{\infty} A_k$. As the lower bound for the sequence of sets $\{A_k\}_{k \geq n}$ is also a set, which belongs to any set in this sequence. It means, we firstly find the belonging relationship through intersection. Then we can compare the size of sets. And we know inferior is the largest lower bound, so here the inferior is $\bigcap_{k=n}^{\infty} A_k$.

Remark-3: Also consider a sequence of sets $\{A_k\}_{k \geq n}$, we can say the $\sup_{k \geq n} A_k$ is $\bigcup_{k=n}^{\infty} A_k$. As the upper bound for the sequence of sets $\{A_k\}_{k \geq n}$ is also a set, which includes any set in this sequence. It means, we firstly find the belonging relationship through union. Then we can compare the size of sets. And we know superior is the least upper bound, so here the superior is $\bigcup_{k=n}^{\infty} A_k$.

Lemma 1.3.1. (A Probability Path by Sidney I. Resnick)

Let $\{A_n\}$ be a sequence of subsets of Ω .

(a) For $\limsup_{n \rightarrow \infty}$ we have the interpretation

$$\begin{aligned} \limsup_{n \rightarrow \infty} A_n &= \left\{ \omega : \sum_{n=1}^{\infty} \mathbf{1}_{A_n}(\omega) = \infty \right\} \\ &= \left\{ \omega : \omega \in A_{n_k}, k = 1, 2, \dots \right\} \end{aligned}$$

for some subsequence n_k depending on ω . Consequently, we write

$$\limsup_{n \rightarrow \infty} A_n = [A_n \text{ i.o.}]$$

where *i.o.* stands for infinitely often.

(b) For $\liminf_{n \rightarrow \infty}$ we have the interpretation

$$\begin{aligned} \liminf_{n \rightarrow \infty} A_n &= \left\{ \omega : \omega \in A_n \text{ for all } n \text{ except a finite number} \right\} \\ &= \left\{ \omega : \sum_n \mathbf{1}_{A_n^c}(\omega) < \infty \right\} \\ &= \left\{ \omega : \omega \in A_n, \forall n \geq n_0(\omega) \right\}. \end{aligned}$$

Proof (a)

Firstly, it's easy to show that $\left\{ \omega : \sum_{n=1}^{\infty} \mathbf{1}_{A_n}(\omega) = \infty \right\} = \left\{ \omega : \omega \in A_{n_k}, k = 1, 2, \dots \right\}$.

Secondly, if $\omega \in \limsup_{n \rightarrow \infty} A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k$, then it means for every n , $\omega \in \bigcup_{k=n}^{\infty} A_k$. And so for all n , there exists some $n_k \geq n$ s.t $\omega \in A_{n_k}$, and therefore

$$\sum_{n=1}^{\infty} \mathbf{1}_{A_n}(\omega) \geq \sum_n \mathbf{1}_{A_{n_k}}(\omega) = \infty,$$

which implies

$$\omega \in \left\{ \omega : \sum_{n=1}^{\infty} \mathbf{1}_{A_n}(\omega) = \infty \right\};$$

thus

$$\limsup_{n \rightarrow \infty} A_n \subset \left\{ \omega : \sum_{n=1}^{\infty} \mathbf{1}_{A_n}(\omega) = \infty \right\}.$$

Conversely, if

$$\omega \in \left\{ \omega : \sum_{n=1}^{\infty} \mathbf{1}_{A_n}(\omega) = \infty \right\},$$

$\sum_{n=1}^{\infty} \mathbf{1}_{A_n}(\omega) = \infty$ means: fixed ω , we have infinitely set A_{n_k} s.t. $\omega \in A_{n_k}$. So, it means there exists $n_k \rightarrow \infty$ s.t. $\omega \in A_{n_k}$, and therefore for all n , $\omega \in \bigcup_{k=n}^{\infty} A_k$ so that $\omega \in \limsup_{n \rightarrow \infty} A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k$. By definition

$$\left\{ \omega : \sum_{j=1}^{\infty} \mathbf{1}_{A_j}(\omega) = \infty \right\} \subset \limsup_{n \rightarrow \infty} A_n.$$

Proof (b)

Firstly, it's easy to show that

$$\begin{aligned} \left\{ \omega : \omega \in A_n \text{ for all } n \text{ except a finite number} \right\} &= \left\{ \omega : \sum_n \mathbf{1}_{A_n^c}(\omega) < \infty \right\} \\ &= \left\{ \omega : \omega \in A_n, \forall n \geq n_0(\omega) \right\}. \end{aligned}$$

Secondly, if $\omega \in \liminf_{n \rightarrow \infty} A_n = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k$, then it means, for fixed ω , there exist n_0 s.t. $\forall k \geq n_0, \omega \in \bigcap_{k=n}^{\infty} A_k$. So, we can get $\forall k \geq n_0, \omega \in A_{n_k}$. It implies

$$\omega \in \left\{ \omega : \omega \in A_n, \forall n \geq n_0(\omega) \right\};$$

thus

$$\liminf_{n \rightarrow \infty} A_n \subset \left\{ \omega : \omega \in A_n, \forall n \geq n_0(\omega) \right\}$$

Conversely, same logic in (a), no illustration.

3 Borel-Cantelli Lemma

For a sequence of random variables $\{X_n, n \geq 0\}$, suppose we need to show $X_n \rightarrow X_0$ almost surely. As we will see, this means that we need to show

$$P \left\{ \omega : \lim_{n \rightarrow \infty} X_n(\omega) = X_0(\omega) \right\} = 1.$$

We will show later that a criterion for this is that for all $\varepsilon > 0$

$$P([|X_n - X_0| > \varepsilon] \text{ i.o.}) = 0.$$

That is, with $A_n = [|X_n - X_0| > \varepsilon]$, we need to check

$$P\left(\limsup_{n \rightarrow \infty} A_n\right) = 0.$$

The Borel-Cantelli Lemma is very simple but still is the basic tool for proving almost sure convergence.

Borel–Cantelli Lemma Let $\{A_n\}$ be any events. If

$$\sum_n P(A_n) < \infty,$$

then

$$P([A_n \text{ i.o.}]) = P\left(\limsup_{n \rightarrow \infty} A_n\right) = 0.$$

Proof. We have

$$\begin{aligned} P([A_n \text{ i.o.}]) &= P\left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k\right) = P\left(\lim_{n \rightarrow \infty} \bigcup_{j \geq n} A_j\right) \\ &= \lim_{n \rightarrow \infty} P\left(\bigcup_{j \geq n} A_j\right) \quad (\text{continuity of } P) \\ &\leq \limsup_{n \rightarrow \infty} \sum_{j=n}^{\infty} P(A_j) \quad (\text{subadditivity}) \\ &= 0, \end{aligned}$$

since $\sum_n P(A_n) < \infty$ implies $\sum_{j=n}^{\infty} P(A_j) \rightarrow 0$, as $n \rightarrow \infty$.